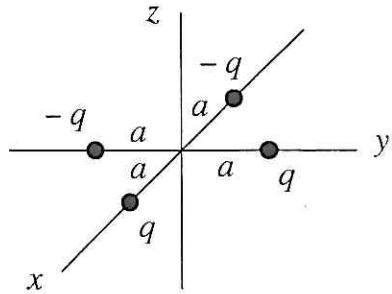
4.1 Calculate the multipole moments q_{lm} of the charge distributions shown as parts a) and b). Try to obtain results for the nonvanishing moments valid for all l, but in each case find the first *two* sets of nonvanishing moments at the very least.

a)



The multipole moments are given by

$$q_{lm} = \int \rho(\vec{x}) r^l Y^*_{lm}(\theta,\phi)\, d^3x = qa^l[Y^*_{lm}(\tfrac{\pi}{2},0)+Y^*_{lm}(\tfrac{\pi}{2},\tfrac{\pi}{2})-Y^*_{lm}(\tfrac{\pi}{2},\pi)-Y^*_{lm}(\tfrac{\pi}{2},\tfrac{3\pi}{2})]$$

This is given in terms of associated Legendre polynomials by

$$q_{lm} = qa^l\sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}P_l^m(0)[1+(-i)^m-(-1)^m-(i)^m]$$

The moments vanish unless m is odd. Writing $m = 2k+1$ gives

$$\begin{aligned} q_{l,2k+1} &= 2qa^l[1-i(-1)^k]\sqrt{\frac{2l+1}{4\pi}\frac{(l-(2k+1))!}{(l+(2k+1))!}}P_l^{2k+1}(0) \\ &= 2qa^l[1-i(-1)^k]Y_{l,2k+1}(\tfrac{\pi}{2},0) \end{aligned}$$

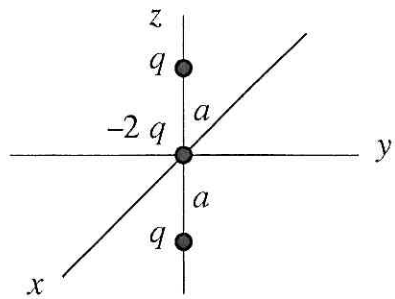
Note that by parity this vanishes unless l is odd. Hence only the odd l and m moments are present. The lowest non-trivial ones are

$$q_{1,1} = -q^*_{1,-1} = -2qa(1-i)\sqrt{\frac{3}{8\pi}}$$

and

$$q_{3,3} = -q^*_{3,-3} = -2qa^3(1+i)\frac{1}{4}\sqrt{\frac{35}{4\pi}} \qquad q_{3,1} = -q^*_{3,-1} = 2qa^3(1-i)\frac{1}{4}\sqrt{\frac{21}{4\pi}}$$

b)



In this case, we have

$$q_{lm} = qa^l[Y^*_{lm}(0,0) + Y^*_{lm}(\pi,0)]$$

for $l > 0$ and $q_{00} = 0$. By azimuthal symmetry, only the $m = 0$ moments are non-vanishing. Hence

$$q_{l0} = qa^l\sqrt{\frac{2l+1}{4\pi}}[P_l(1) + P_l(-1)] = qa^l[1 + (-1)^l]\sqrt{\frac{2l+1}{4\pi}} \qquad l > 0$$

We end up with even multipoles

$$q_{l0} = qa^l\sqrt{\frac{2l+1}{\pi}} \qquad l = 2, 4, 6, \ldots$$

Explicitly

$$q_{20} = qa^2\sqrt{\frac{5}{\pi}} \qquad q_{40} = qa^4\sqrt{\frac{9}{\pi}}$$

c) For the charge distribution of the second set b) write down the multipole expansion for the potential. Keeping only the lowest-order term in the expansion, plot the potential in the x-y plane as a function of distance from the origin for distances greater than a.

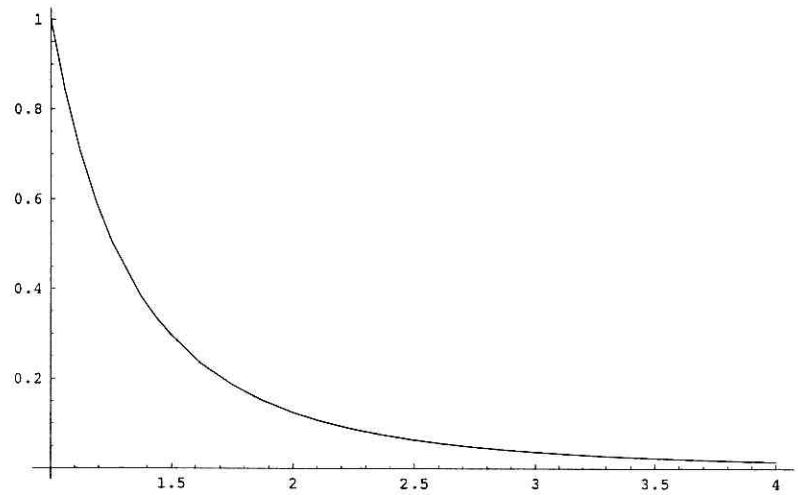
The expansion of the potential is

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0}\sum_{l,m}\frac{4\pi}{2l+1}q_{lm}\frac{Y_{lm}(\theta,\phi)}{r^{l+1}} = \frac{1}{\epsilon_0}\sum_{l=2,4,\ldots}\frac{qa^l}{2l+1}\sqrt{\frac{2l+1}{\pi}}\frac{Y_{l0}(\theta,\phi)}{r^{l+1}}$$
$$= \frac{q}{2\pi\epsilon_0}\sum_{l=2,4,\ldots}\frac{a^l}{r^{l+1}}P_l(\cos\theta) = \frac{q}{4\pi\epsilon_0}\frac{a^2}{r^3}(3\cos^2\theta - 1) + \cdots$$

In the x-y plane we have $\cos\theta = 0$, so the lowest order term is

$$\Phi = -\frac{q}{4\pi\epsilon_0 a}\left(\frac{a}{r}\right)^3 + \cdots$$

We all know what $1/r^3$ looks like when plotted, but here it is

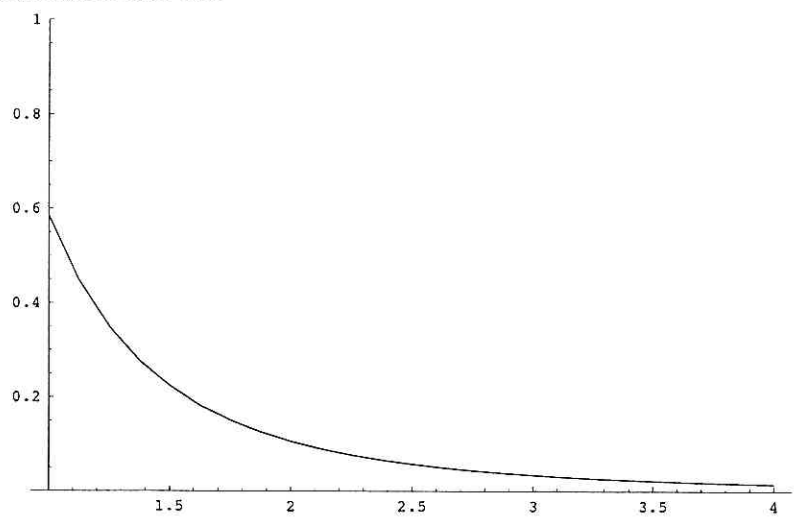


d) Calculate directly from Coulomb's law the exact potential for *b*) in the x-y plane. Plot it as a function of distance and compare with the result found in part *c*).

For three charges, the potential is simply the sum of three terms, one for each charge. In the x-y plane, if r is the distance from the origin we have

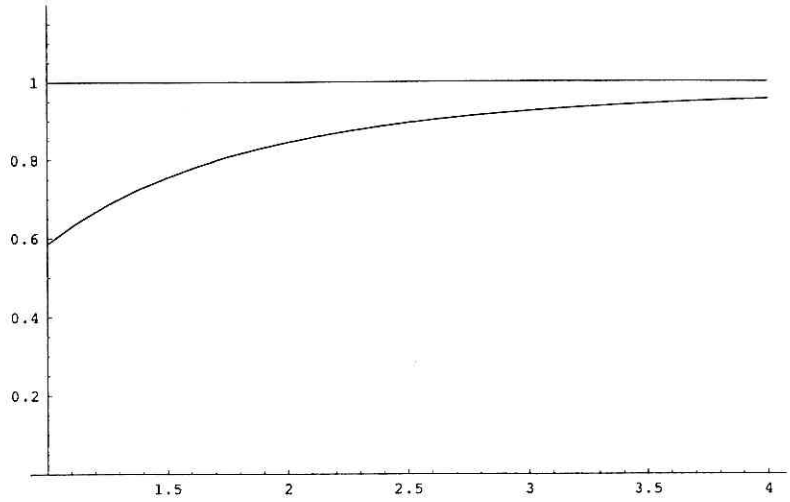
$$\Phi = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{\sqrt{r^2+a^2}} - \frac{1}{r} + \frac{1}{\sqrt{r^2+a^2}}\right) = -\frac{q}{2\pi\epsilon_0 r}\left(1 - \frac{1}{\sqrt{1+(a/r)^2}}\right)$$
$$= -\frac{q}{4\pi\epsilon_0 a}2\left(\frac{1}{(r/a)} - \frac{1}{\sqrt{1+(r/a)^2}}\right)$$

The exact potential looks like



Divide out the asymptotic form in parts *c*) and *d*) to see the behavior at large distances more clearly.

If we divide out by $1/r^3$, the approximate and exact potentials are



where the straight line is the approximation of c) and the sloped line is the exact result. The approximation improves as $r \gg a$.